Two pictures of one top hat

The shell equation, the nonlinear equation for δ, and the mean density of our seed

A companion note to *From a primordial fluctuation to an idealised rotation curve*

B. R. Parodi

24 October 2026; extended 29 October 2026

Abstract

The collapse of a uniform overdense sphere, the *top hat*, can be described in two ways: by Newton's equation for the radius of its edge, or by a nonlinear differential equation for its density contrast δ, the form used in most papers on spherical collapse in dark-energy cosmologies. We show step by step that the two are the same equation, that the cosmological constant drops out of the equation for δ because it acts on the sphere exactly as on the background, and that the cycloid of our main paper is the exact solution of the δ equation in an Einstein–de Sitter universe. The mean overdensity inside each shell of our seed is therefore a solution of the nonlinear δ equation, one for every shell: the seed is a family of nested top hats, which share one solution shifted in time. We give the closed form, its linear limit and its values at turnaround ($1+\langle\delta\rangle = 5.55$), halting (147) and collapse (∞), verify the equivalence numerically in a flat ΛCDM background to better than 10^{-6}, and state where the δ picture ends: at shell crossing, which in our model begins at the halting front. Figures of the nested top hats at one and at two moments show how a common energy, together with the halting rule, builds the flat curve behind the accretion front at $R_b = 0.175\,vt$, and what the density profile of the seed looks like inside and outside the front.

1 Why two pictures?

A spherical region that is slightly denser than its surroundings expands more slowly than the Universe, stops, and falls back. If the region is uniform, a top hat, it behaves as a small closed universe of its own (Gunn & Gott, 1972; Peebles, 1980). Two descriptions are in use:

- the **shell (or radius) picture**: follow the radius $R(t)$ of the edge of the sphere, which encloses a fixed mass M and obeys Newton's law. This is what our main paper does, shell by shell;
- the **density picture**: follow the density contrast $\delta(t) = \rho/\bar\rho - 1$ of the sphere relative to the background. This is the language of cosmological perturbation theory, and the one in which spherical collapse is extended to dark energy, modified gravity, shear and rotation (e.g. Fosalba & Gaztañaga, 1998; Del Popolo et al., 2013; Herrera et al., 2019).

The second picture produces an equation that looks complicated,

$$\ddot\delta + 2H\dot\delta - \frac{4}{3}\frac{\dot\delta^2}{1+\delta} = 4\pi G\bar\rho\,\delta(1+\delta) = \frac{3}{2}H^2\Omega_{\rm m}(t)\,\delta(1+\delta), \tag{1}$$

where $H = \dot a/a$ is the Hubble rate, $\bar\rho$ the mean matter density of the background and $\Omega_{\rm m}(t) = 8\pi G\bar\rho/3H^2$. It is eq. (10) of Herrera et al. (2019) for a cosmological constant (their $\gamma = 1$, $\delta_{\rm de} = 0$), and the starting point of many papers. The aim of this note is to show that Eq. (1) contains nothing that is not already in the shell equation of our paper, and to see what this means for our seed.

2 The shell picture

A shell of mass M at radius R is pulled by the mass inside it and, in the presence of a cosmological constant Λ (in units of $1/\text{time}^2$), pushed outward by $\Lambda R/3$:

$$\ddot{R} = -\frac{GM}{R^2} + \frac{\Lambda}{3} R. \tag{2}$$

As long as no other shell crosses it, M is constant. Multiplying by $\dot{R}$ and integrating gives the conserved specific energy

$$E = \frac{1}{2}\dot{R}^2 - \frac{GM}{R} - \frac{\Lambda}{6}R^2. \tag{3}$$

For $\Lambda = 0$ and $E < 0$ the solution is the cycloid

$$R = A(1 - \cos\eta), \qquad t = B(\eta - \sin\eta), \qquad A = \frac{GM}{2|E|}, \quad B = \frac{GM}{(2|E|)^{3/2}}, \tag{4}$$

which reaches its maximum, *turnaround*, $R_{\rm ta} = 2A$, at $\eta = \pi$, and collapses to a point at $\eta = 2\pi$. Note the identity

$$\frac{A^3}{B^2} = GM, \tag{5}$$

which follows directly from the definitions of A and B and will make the mass drop out below.

The background obeys the same kind of equation. A sphere of the *unperturbed* Universe of comoving radius X has physical radius aX, and the Friedmann equations for matter and Λ read

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\bar{\rho} + \frac{\Lambda}{3}, \qquad H^2 = \frac{8\pi G}{3}\bar{\rho} + \frac{\Lambda}{3}, \qquad \bar{\rho}\, a^3 = \text{const}. \tag{6}$$

In words: the background is itself a family of shells, each obeying Eq. (2) with the mean density inside it. The top hat differs from the background only in that it contains more mass for its size.

3 From the shell to the density contrast

The mean density inside the shell is $\rho = 3M/(4\pi R^3)$. Divided by the background density, and using $\bar{\rho}a^3 = \text{const}$,

$$1 + \delta = \frac{\rho}{\bar{\rho}} = K\left(\frac{a}{R}\right)^3, \qquad K = \text{const}. \tag{7}$$

The density contrast is nothing but the cube of the ratio of two radii: that of a background sphere, which grows as a, and that of the shell, R. Everything else follows by differentiating twice.

First derivative. Take the logarithm of Eq. (7), $\ln(1+\delta) = \ln K + 3\ln a - 3\ln R$, and differentiate with respect to time. With $h \equiv \dot{R}/R$, the local expansion rate of the shell,

$$\frac{\dot{\delta}}{1+\delta} = 3\,(H - h) \qquad \Longleftrightarrow \qquad h = H - \frac{\dot{\delta}}{3(1+\delta)}. \tag{8}$$

The contrast grows exactly when the shell expands more slowly than the background. This is also the continuity equation of a uniform fluid sphere: $\dot{\rho}/\rho = -3h$ and $\dot{\bar{\rho}}/\bar{\rho} = -3H$.

Second derivative. Differentiate Eq. (8) once more:

$$\frac{\ddot{\delta}}{1+\delta} - \frac{\dot{\delta}^2}{(1+\delta)^2} = 3\,(\dot{H} - \dot{h}). \tag{9}$$

For any scale factor, $\dot{H} = \ddot{a}/a - H^2$, and likewise $\dot{h} = \ddot{R}/R - h^2$. Hence

$$\dot{H} - \dot{h} = \left(\frac{\ddot{a}}{a} - \frac{\ddot{R}}{R}\right) - (H^2 - h^2). \tag{10}$$

The first bracket contains the forces, the second the kinematics.

The forces: Λ cancels. From Eqs. (2) and (6),

$$\frac{\ddot{a}}{a} - \frac{\ddot{R}}{R} = \left(-\frac{4\pi G}{3}\bar{\rho} + \frac{\Lambda}{3}\right) - \left(-\frac{GM}{R^3} + \frac{\Lambda}{3}\right) = \frac{4\pi G}{3}\,(\rho - \bar{\rho}) = \frac{4\pi G}{3}\,\bar{\rho}\,\delta. \tag{11}$$

This is the heart of the matter. The cosmological constant produces an acceleration proportional to distance, $\Lambda R/3$, so it stretches the shell and the background at exactly the same rate and drops out of their *difference*. Only the excess of matter, $\rho - \bar{\rho}$, drives the contrast. This is why Λ "cannot cluster": in the language of Herrera et al. (2019), the cosmological constant has $\delta_{\rm de} = 0$ for every top hat. It still enters Eq. (1), but only through the background, i.e. through $H(t)$ and $\Omega_{\rm m}(t)$.

The kinematics. With Eq. (8), $H - h = \dot{\delta}/[3(1+\delta)]$ and $H + h = 2H - \dot{\delta}/[3(1+\delta)]$, so

$$3\,(H^2 - h^2) = 3(H-h)(H+h) = \frac{2H\dot{\delta}}{1+\delta} - \frac{\dot{\delta}^2}{3(1+\delta)^2}. \tag{12}$$

Putting it together. Inserting Eqs. (11) and (12) into Eqs. (10) and (9),

$$\frac{\ddot{\delta}}{1+\delta} - \frac{\dot{\delta}^2}{(1+\delta)^2} = 4\pi G\bar{\rho}\,\delta - \frac{2H\dot{\delta}}{1+\delta} + \frac{\dot{\delta}^2}{3(1+\delta)^2},$$

and multiplying by $1+\delta$ gives exactly Eq. (1):

$$\ddot{\delta} + 2H\dot{\delta} - \frac{4}{3}\frac{\dot{\delta}^2}{1+\delta} = 4\pi G\bar{\rho}\,\delta(1+\delta).$$

Each term now has a plain meaning. $2H\dot{\delta}$ is the *Hubble drag*: the expanding background dilutes the growth. The term $\frac{4}{3}\dot{\delta}^2/(1+\delta)$ is purely kinematic; it comes from writing a ratio of volumes, Eq. (7), as a contrast, and it is what makes the growth run away once $\delta \gtrsim 1$. The right-hand side is the gravitational pull of the excess mass, $\bar{\rho}\delta$, enhanced by the factor $1+\delta$ because the sphere has shrunk relative to the background.

The fluid route, and where shear and rotation enter. The same equation follows from the continuity, Euler and Poisson equations of a pressureless fluid. Taking the divergence of the Euler equation gives the Raychaudhuri equation for the expansion $\theta = \nabla\cdot\mathbf{u}$, which contains the shear σ^2 and the vorticity ω^2 of the flow; the result is Eq. (1) with an extra term $-(1+\delta)(\sigma^2-\omega^2)$ on the left (Fosalba & Gaztañaga, 1998; Del Popolo et al., 2013). For a top hat both vanish: a uniform sphere expanding or contracting radially is neither sheared nor rotated. Models that keep $\sigma^2 - \omega^2$ go beyond the top hat; in the shell picture the rotation corresponds to a centrifugal term j^2/R^3 in Eq. (2), which our main paper adds to each shell only after its turnaround.

4 The exact solution: the cycloid

Without Λ, in an Einstein–de Sitter (EdS) background, $\bar{\rho} = 1/(6\pi G t^2)$, $H = 2/(3t)$ and $\Omega_{\rm m} = 1$. With the cycloid (4) and the identity (5),

$$1 + \langle\delta\rangle = \frac{3M}{4\pi A^3(1-\cos\eta)^3}\,6\pi G B^2(\eta - \sin\eta)^2 = \frac{9}{2}\,\frac{GMB^2}{A^3}\,\frac{(\eta-\sin\eta)^2}{(1-\cos\eta)^3} = \frac{9}{2}\,\frac{(\eta-\sin\eta)^2}{(1-\cos\eta)^3}. \tag{13}$$

We write $\langle\delta\rangle$ for the mean overdensity inside a shell, as in the main paper. Two remarks:

- **The mass and the energy have dropped out.** Every top hat in EdS runs through the same sequence of $1+\langle\delta\rangle$ as a function of its phase η; only the clock, $t = B(\eta - \sin\eta)$, depends on M and E. Equation (1) in EdS has no scale, so one solution serves all top hats.
- **It is an exact solution.** Equation (13), read as a function of t through $t = B(\eta - \sin\eta)$, solves Eq. (1) identically. Fig. 1a confirms this numerically: integrating Eq. (1) from initial values taken from the cycloid at $\eta = 0.05$ reproduces Eq. (13) to 2×10^{-8} up to $\eta = 1.9\pi$.

The linear limit. For small η, $\eta - \sin\eta = \frac{\eta^3}{6}(1 - \frac{\eta^2}{20} + \dots)$ and $1 - \cos\eta = \frac{\eta^2}{2}(1 - \frac{\eta^2}{12} + \dots)$, so

$$1 + \langle\delta\rangle = 1 + \frac{3}{20}\eta^2 + O(\eta^4), \qquad \langle\delta\rangle \simeq \frac{3}{20}\Big(\frac{6t}{B}\Big)^{2/3} \propto t^{2/3} \propto a. \tag{14}$$

This is the growing mode of linear theory. Indeed, dropping the quadratic terms of Eq. (1) in EdS leaves $\ddot\delta + \frac{4}{3t}\dot\delta - \frac{2}{3t^2}\delta = 0$, whose solutions are $\delta \propto t^{2/3}$ (growing) and $\delta \propto t^{-1}$ (decaying). The cycloid starts on the pure growing mode, the standard choice; the growing mode solution of Herrera et al. (2019, their eq. 19), $\delta_\mathrm{m} = Ca$, is the same thing.

Milestones. Table 1 lists the values at the phases that matter for our model, together with the *linear* contrast δ_lin, i.e. Eq. (14) extrapolated to the same time. The linear value at collapse, $\delta_\mathrm{c} = 1.686$, is the familiar threshold of the Press–Schechter theory.

Table 1: Milestones of a top hat in EdS.

	phase η	t/t_ta	$1 + \langle\delta\rangle$	δ_lin
turnaround	π	1	$9\pi^2/16 = 5.55$	1.062
halting (front)	$3\pi/2$	$(1 + 3\pi/2)/\pi = 1.82$	$\frac{9}{2}(1 + 3\pi/2)^2 = 146.8$	1.583
collapse	2π	2	∞	1.686
“virialised” (convention)	–	2	$18\pi^2 = 177.7$	1.686

The last two rows deserve a comment. In the standard top-hat model the sphere collapses to a point at $t = 2t_\mathrm{ta}$, which is unphysical; by convention one assumes it has virialised at half its turnaround radius by then, which gives $18\pi^2 \approx 178$. Our shells reach the same radius, $R_\mathrm{ta}/2$, earlier, at $\eta = 3\pi/2$ on their way back ($1 - \cos\frac{3\pi}{2} = 1$), and halt there. Same radius, earlier time: the background is denser at $1.82\,t_\mathrm{ta}$ than at $2t_\mathrm{ta}$ by $(2/1.82)^2 = 1.21$, which is exactly the ratio $177.7/146.8$.

5 With a cosmological constant

With $\Lambda > 0$ neither Eq. (2) nor Eq. (1) has a closed-form solution, but they remain the same equation. To check this numerically, we integrate both in a flat ΛCDM background with $\Omega_\mathrm{m} = 0.3$ (units $H_0 = 1$, so $\Lambda = 3\Omega_\Lambda H_0^2 = 2.1$), from $a_\mathrm{i} = 1.9 \times 10^{-3}$ on the growing mode, for three initial amplitudes (Fig. 1b): the two descriptions agree to better than 4×10^{-7} along the whole evolution.

The turnaround threshold. The energy (3) shows what Λ changes. The potential $-GM/R - \Lambda R^2/6$ has a maximum at the radius $R_\Lambda = (3GM/\Lambda)^{1/3}$, the *maximum turnaround radius* (Pavlidou & Tomaras, 2014). A shell turns around only if its energy lies below the top of this barrier,

$$E \le -\frac{3}{2}\,(GM)^{2/3}\Big(\frac{\Lambda}{3}\Big)^{1/3}; \tag{15}$$

for our seed, $E = -v^2/2$, this is the condition $v^2 \ge 3(GM)^{2/3}(\Lambda/3)^{1/3}$ of the main paper (Sect. 3.2), which bounds the mass of the seed. In the density picture the same threshold appears differently: a top hat just above it collapses, and its δ runs away; one just below it never turns around, and its δ *freezes* at a finite value ($1 + \delta \to 23.6$ for the example in Fig. 1b), because at late times shell and background both expand exponentially at the same rate $\sqrt{\Lambda/3}$, so that a/R in Eq. (7) tends to a constant. This is the “Λ stops structure formation” of cosmology textbooks, seen from the two sides.

Dark energy that clusters. Herrera et al. (2019) go further and let the dark energy inside the sphere differ from the background (their parameters γ and c_s^2). Then it no longer cancels in Eq. (11), a second contrast $\delta_{\rm de}$ appears on the right-hand side of Eq. (1), and energy inside the sphere is not conserved. For a cosmological constant, the case of our model, none of this arises.

6 Our seed: a family of nested top hats

Our seed gives every shell the same specific energy, $E = -v^2/2$. With Eq. (4), $A = GM/v^2$ and $B = GM/v^3$, so the clock of each shell runs in proportion to its mass:

$$t = \frac{GM}{v^3}\,(\eta - \sin\eta), \qquad t_{\rm ta} = \pi\,\frac{GM}{v^3} \propto M. \tag{16}$$

Every shell is the edge of a top hat of its own mass (the main paper calls the seed a sequence of *nested top hats*), and as long as no shells cross, the mass inside each shell is constant and Eq. (2) holds for it. Therefore:

1. **The mean overdensity inside every shell obeys Eq.** (1), each shell with its own initial amplitude.

2. **All shells share one solution, Eq.** (13), shifted in time. At a given time t the shell of mass M is at the phase $\eta(M,t)$ given by Eq. (16), so

$$1 + \langle\delta\rangle(M,t) = \frac{9}{2}\,\frac{(\eta-\sin\eta)^2}{(1-\cos\eta)^3}, \qquad \eta - \sin\eta = \frac{v^3 t}{GM}. \tag{17}$$

   The mean overdensity depends on M and t only through t/M: this is the self-similarity of the seed, seen in the density picture.

3. **The initial profile is $\langle\delta\rangle \propto M^{-2/3}$.** At early times Eq. (14) gives $\langle\delta\rangle \simeq \frac{3}{20}(6v^3t/GM)^{2/3} \propto t^{2/3}M^{-2/3}$: the growing mode with an amplitude that falls with mass as $M^{-2/3}$, the statement of Sect. 3.2 of the main paper (Fig. 1c, dashed). Inner shells are denser, reach every milestone earlier, and turn around first.

4. **The δ picture ends at the front.** The equation for a top hat holds only while the mass inside the shell is constant. In our model the shells do not cross before they halt: a shell halts at $R_b \simeq GM/v^2$, which grows with M, so inner shells stop inside outer ones. At the halting phase $\eta_b = 3\pi/2$ every shell has $1 + \langle\delta\rangle = 146.8$, and the shells with $M < v^3t/[G(1+3\pi/2)]$ have halted (shaded in Fig. 1c). Inside the front the mass distribution is the settled galaxy of the main paper, and its mean density is no longer given by Eq. (17); in a collisionless system shells would cross there, and the top-hat equation would fail for the same reason.

7 The nested top hats in pictures

Figures 2 and 3 (units $G = v = 1$) show the seed of Sect. 6 in three ways and at two moments.

One history, stretched (Fig. 2a). Each curve has the same shape, stretched by M in both radius and time. A shell expands with the Hubble flow, slows down under the pull of its own mass, turns around at $R_{\rm ta} = 2GM/v^2$ and $t_{\rm ta} = \pi GM/v^3$ (dots), and falls back. A shell twice as heavy goes twice as far and takes twice as long. In the main paper a shell stops falling where the galaxy already built encloses its mass (its assumption A3). For a galaxy with the flat speed v this happens at about half the turnaround radius, $R \simeq GM/v^2$, i.e. at the phase $\eta_b \simeq 3\pi/2$,

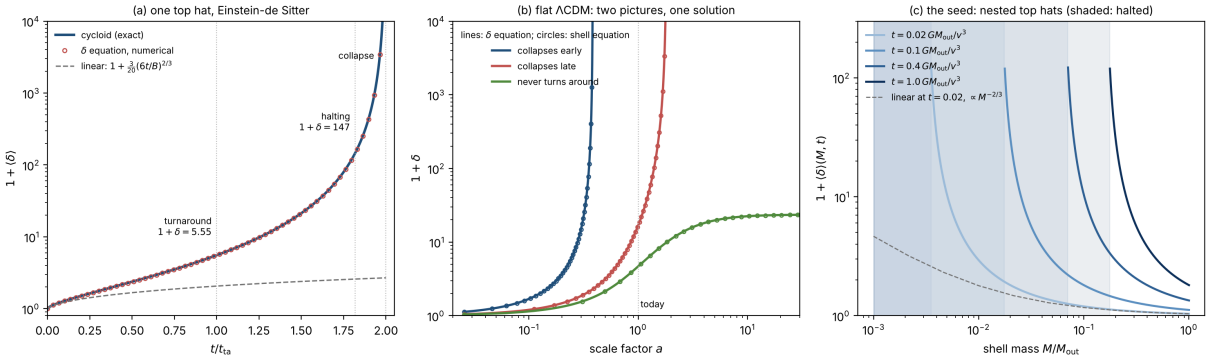


Figure 1: (a) One top hat in an Einstein–de Sitter universe: the cycloid, Eq. (13) (line), and the numerical solution of the δ equation, Eq. (1) (circles), with the linear growth (dashed). Vertical lines mark turnaround, halting and collapse. (b) Three top hats in flat ΛCDM ($\Omega_\mathrm{m} = 0.3$), with initial amplitudes 3, 1.15 and 0.85 times the turnaround threshold, Eq. (15): lines from the δ equation, circles from the shell equation, Eq. (2), converted with Eq. (7). The shell below the threshold never turns around and its contrast freezes. (c) The seed of the main paper as nested top hats: $1 + \langle\delta\rangle$ against shell mass at four times, Eq. (17); each curve ends at the halting front, $1 + \langle\delta\rangle = 147$, and the shaded regions hold the halted shells, where the top-hat description no longer applies. The dashed line is the linear profile $\langle\delta\rangle \propto M^{-2/3}$ at the earliest time.

three quarters of the way along the cycloid. Since this halting phase is the same for every shell, the halting points lie on a straight line through the origin, the *accretion front*

$$R_b = \frac{v\,t}{1 + 3\pi/2} \simeq 0.175\,vt.$$

Behind the front the halted mass grows in proportion to the radius, $M \propto R_b$, so that the circular speed $v_c^2 = GM/R$ is constant: a common energy, together with the halting rule, builds a flat rotation curve.

The nested spheres at one moment (Fig. 2b). At a fixed time the shells are at different stages of the same history. The lightest ones started their collapse earliest and have already halted; they form the settled galaxy inside the front (grey). Somewhat heavier shells have turned around and are falling back towards the front. The heaviest ones are still expanding, though more slowly than the background. At a later time the whole picture is the same, only larger: this is the self-similarity of the growth (main paper, Sect. 5.1), in which every radius scales with vt.

Why it is one top hat (Fig. 2c). The mean overdensity inside every shell follows the same curve, Eq. (17). Early on it grows as the linear growing mode, $\langle\delta\rangle \propto t^{2/3}M^{-2/3}$: at a given time the inner (lighter) shells are denser, which is why they turn around first and why shells do not overtake each other before they halt. At turnaround every shell has $1 + \langle\delta\rangle = 5.55$, and when it halts $1 + \langle\delta\rangle = 147$ (Table 1). In the figure the curves for different masses are identical copies shifted along the time axis by the factor M (the axis is logarithmic, so equal factors in M give equal shifts).

Two moments, and the density profile (Fig. 3). The same seed, now extended to $M = 4$, at $t = 4$ and $t = 8$, with the density profile at each time. The density is computed from the shells themselves: at radius R it is $\rho = (\partial M/\partial R)_t/4\pi R^2$, the mass per unit radius divided by the area of the sphere.

- *Inside the front*, the halted matter has $M \propto R$, i.e. the density of a singular isothermal sphere, $\rho = v^2/(4\pi G R^2)$. (In the main paper this matter settles into the P18 profile, whose

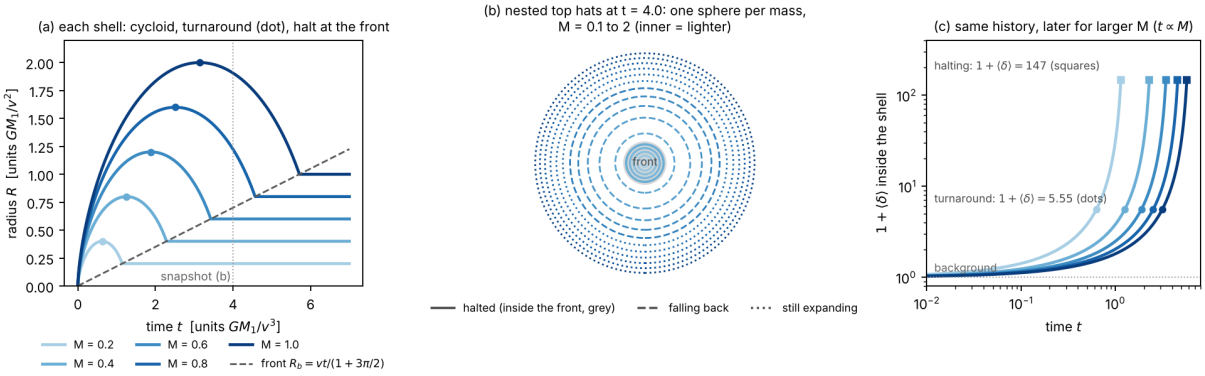


Figure 2: Nested top hats with one common energy $E = -v^2/2$ (units $G = v = 1$; darker blue = larger mass). (a) Radius of five shells, $M = 0.2$ to 1, against time: each follows its cycloid, turns around (dot) and halts on the way back where it meets the accretion front (dashed line). (b) The nested spheres at one moment, $t = 4$, for $M = 0.1$ to 2: inner spheres have halted inside the front (solid, grey disc), the middle ones are falling back (dashed), the outer ones are still expanding (dotted). (c) The mean overdensity inside each shell of panel (a): the same curve for every M, Eq. (17), shifted to later times in proportion to M; dots mark turnaround, squares halting. Script: `code/nested.py`.

core of a few per cent of R_b is omitted here.) Because v is fixed, this part of the profile is *the same at both times*: the galaxy does not change where it already exists, it only grows outwards as the front advances.

- *At the front* the density drops by a factor $D \simeq 6.7$ (main paper, Sect. 5.1 and Appendix B): just outside, the shells are falling back and are spread out in radius, whereas just inside, the halted shells are packed at their final radii.

- *Outside the front*, the infalling shells (dashed) and, beyond the current turnaround radius $R_{\rm ta} = 2vt/\pi$, the still expanding ones (dotted) have a density that falls slowly outwards towards that of the background.

- *Beyond the edge of the seed* (here the shell $M = 4$) lies the unperturbed background, $\bar\rho = 1/(6\pi G t^2)$ (grey).

From $t = 4$ to $t = 8$ the front and the turnaround radius move out by a factor of two, and the density outside the front drops by a factor of four ($\propto t^{-2}$), in step with the background. In the variable $\xi = R/vt$ the two outer profiles would lie exactly on top of each other: this is the self-similar function $P(\xi)$ of the main paper (Sect. 5.1, Fig. 2).

Why the shells halt at the front at all, and what the halted matter must do to settle into the observed profile, are the subject of the companion note *Halting and settling*.

8 Summary

- The nonlinear equation for the density contrast of a top hat, Eq. (1), is Newton's equation for the radius of its edge, Eq. (2), rewritten with $1+\delta \propto (a/R)^3$ (Sect. 3). The cosmological constant cancels between shell and background and enters only through $H(t)$ and $\Omega_{\rm m}(t)$.

- In EdS the cycloid is its exact solution, Eq. (13), the same for every mass and energy; its linear limit is the growing mode $\langle\delta\rangle \propto a$.

- The mean density of our seed is this solution, one copy per shell, shifted in time by $t_{\rm ta} \propto M$: the seed is a self-similar family of nested top hats, with $1 + \langle\delta\rangle = 5.55$ at turnaround and 147 at the halting front.
- With the halting rule, the nested top hats build a flat rotation curve behind the front $R_b = 0.175\, vt$; the density is that of a singular isothermal sphere inside the front, drops by a factor of 6.7 across it, and outside it falls with time in step with the background (Sect. 7).
- The density picture holds only until shells cross; in our model that is where the front halts them. Extensions of Eq. (1) to clustering dark energy, shear or vorticity (Del Popolo et al., 2013; Herrera et al., 2019) add physics that a top hat in ΛCDM does not have.

References

Del Popolo, A., Pace, F., & Lima, J. A. S. 2013, MNRAS, 430, 628

Fosalba, P., & Gaztañaga, E. 1998, MNRAS, 301, 503

Gunn, J. E., & Gott, J. R. 1972, ApJ, 176, 1

Herrera, D., Waga, I., & Jorás, S. E. 2019, arXiv:1906.04326

Pavlidou, V., & Tomaras, T. N. 2014, JCAP, 09, 020

Peebles, P. J. E. 1980, The Large-Scale Structure of the Universe (Princeton: Princeton University Press)

A Code

The script `code/tophat.py` (numpy, scipy, matplotlib) performs the three checks of this note and draws Fig. 1; its output is in `code/tophat_output.txt`. The scripts `code/nested.py` and `code/nested2.py` draw Figs. 2 and 3.

```
"""Companion note "Top hats and the nonlinear equation for delta".
Checks (i) that the cycloid solves the nonlinear top-hat equation in Einstein-de Sitter,
(ii) that the shell equation with Lambda and the delta equation (Herrera et al. eq. 10, Lambda case)
give the same evolution in a flat LCDM background, and (iii) draws the mean overdensity of the
nested top hats of the pf2irc seed. Units: H0 = 1 (time in 1/H0), G = 1; Lambda in 1/time^2.
python3 tophat.py -> fig_tophat.png, tophat_output.txt"""
import numpy as np
from scipy.integrate import solve_ivp
from scipy.optimize import brentq
import matplotlib; matplotlib.use("Agg"); import matplotlib.pyplot as plt

out = open("tophat_output.txt", "w")
def say(*a):
    s = " ".join(str(x) for x in a); print(s); out.write(s + "\n")

# ---------------------------------------------------------------- (i) Einstein-de Sitter
# cycloid of a top hat: R = A(1-cos eta), t = B(eta - sin eta); 1+delta = 9(eta-sin)^2/(2(1-cos)^3)
def _u(eta): # eta - sin(eta), with its series for small eta (no cancellation)
    eta = np.asarray(eta, float)
    return np.where(eta < 1e-2, eta**3/6 - eta**5/120, eta - np.sin(eta))
def _w(eta): # 1 - cos(eta) = 2 sin^2(eta/2)
    return 2*np.sin(np.asarray(eta, float)/2)**2
def one_plus_delta(eta):
    return 9*_u(eta)**2/(2*_w(eta)**3)
def ddelta_dt(eta): # d delta/dt = (d/d eta)(1+delta) / (dt/d eta), dt/d eta = 1 - cos eta (units of B)
    u, w = _u(eta), _w(eta)
    return 4.5*(2*u*w*w**3 - 3*u**2*w**2*np.sin(eta))/w**6/w
# delta equation in EdS (Omega_m = 1, H = 2/3t, 4 pi G rho_bar = 2/(3 t^2)), time in units of B
def rhs_eds(t, y):
```

```
    d, dd = y
    return [dd, -(4/(3*t))*dd + (4/3)*dd**2/(1+d) + (2/(3*t**2))*d*(1+d)]
eta0 = 0.05
t0 = float(_u(eta0))
d0 = float(one_plus_delta(eta0)) - 1
dd0 = float(ddelta_dt(eta0))
eta_end = 1.9*np.pi
t_end = eta_end - np.sin(eta_end)
sol = solve_ivp(rhs_eds, (t0, t_end), [d0, dd0], rtol=1e-11, atol=1e-14, dense_output=True)
assert sol.success, sol.message
eta_chk = np.linspace(0.3, eta_end, 400)
t_chk = eta_chk - np.sin(eta_chk)
err = np.max(np.abs((1 + sol.sol(t_chk)[0])/one_plus_delta(eta_chk) - 1))
say(f"EdS: max relative difference cycloid vs delta equation up to eta=1.9 pi: {err:.1e}")
for name, eta in [("turnaround", np.pi), ("halting", 1.5*np.pi)]:
    t = eta - np.sin(eta); dlin = 3/20*(6*t)**(2/3)
    say(f" {name:10s} eta={eta/np.pi:.2f} pi: 1+delta = {one_plus_delta(eta):8.3f}, delta_lin = {dlin:.3f}")
say(f" collapse eta=2 pi: 1+delta -> inf, delta_lin = {3/20*(12*np.pi)**(2/3):.3f}")
say(f" halting value (9/2)(1+3pi/2)^2 = {4.5*(1+1.5*np.pi)**2:.2f}; top-hat virial 18 pi^2 = {18*np.pi**2:.1
    f}")

# ------------------------------------------------------------------ (ii) flat LCDM, Lambda in 1/time^2
Om, OL = 0.3, 0.7
Lam = 3*OL # Lambda = 3 H0^2 Omega_Lambda, H0 = 1
def a_of_t(t):
    return (Om/OL)**(1/3)*np.sinh(1.5*np.sqrt(OL)*t)**(2/3)
def H_of_t(t):
    return np.sqrt(OL)/np.tanh(1.5*np.sqrt(OL)*t)
def Om_of_t(t):
    a = a_of_t(t); return Om/a**3/(Om/a**3 + OL)
# shell: comoving radius 1, R_i = a_i (1+d_i)^(-1/3), G M = Om/2 (H0 = 1)
GM = Om/2
def rhs_R(t, y):
    R, V = y; return [V, -GM/R**2 + Lam*R/3]
def rhs_d(t, y): # Herrera et al. (2019) eq. (10) with delta_de = 0 (Lambda)
    d, dd = y; H = H_of_t(t)
    return [dd, -2*H*dd + (4/3)*dd**2/(1+d) + 1.5*H**2*Om_of_t(t)*d*(1+d)]
ti = 1e-4
ai, Hi = a_of_t(ti), H_of_t(ti)
def run(di, tmax=8.0):
    ddi = Hi*di # growing mode, delta ~ a, early matter era
    Ri = ai*(1+di)**(-1/3)
    Vi = Ri*(Hi - ddi/(3*(1+di))) # from delta-dot/(1+delta) = 3(H - Rdot/R)
    stop = lambda t, y: y[0] - 1e-3*ai; stop.terminal = True
    sR = solve_ivp(rhs_R, (ti, tmax), [Ri, Vi], rtol=1e-11, atol=1e-15, dense_output=True, events=stop)
    big = lambda t, y: y[0] - 1e5; big.terminal = True
    sd = solve_ivp(rhs_d, (ti, tmax), [di, ddi], rtol=1e-11, atol=1e-15, dense_output=True, events=big)
    return sR, sd
t0_LCDM = brentq(lambda t: a_of_t(t) - 1, 0.1, 3)
say(f"LCDM (Om=0.3): age t0 = {t0_LCDM:.4f}/H0, Lambda = {Lam:.2f} H0^2")
# threshold: the shell turns around only if (Rdot^2/2 - GM/R - Lam R^2/6) <= -(3/2)(GM)^(2/3)(Lam/3)^(1/3)
def energy(di):
    Ri = ai*(1+di)**(-1/3); Vi = Ri*(Hi - Hi*di/(3*(1+di)))
    return 0.5*Vi**2 - GM/Ri - Lam*Ri**2/6
Ecrit = -1.5*GM**(2/3)*(Lam/3)**(1/3)
dthr = brentq(lambda d: energy(d) - Ecrit, 1e-6, 1e-2)
say(f"turnaround threshold: delta_i = {dthr:.4e} at a_i = {ai:.3e} (delta_i/a_i = {dthr/ai:.4f});"
    f" maximum turnaround radius (3GM/Lambda)^(1/3) = {(3*GM/Lam)**(1/3):.4f} (comoving 1 -> a)")
cases = [("collapses early", 3.0*dthr), ("collapses late", 1.15*dthr), ("never turns around", 0.85*dthr)]
runs = []
for lab, di in cases:
    sR, sd = run(di)
    tt = np.linspace(ti*50, min(sR.t[-1], sd.t[-1])*0.999, 600)
    R = sR.sol(tt)[0]; dR = (a_of_t(tt)/R)**3 - 1
    dD = sd.sol(tt)[0]
    m = dD < 1e4
    rel = np.max(np.abs((1+dR[m])/(1+dD[m]) - 1))
    tta = sR.t[np.argmax(sR.y[0])] if sR.y[0].argmax() < len(sR.t)-1 else np.nan
    say(f" {lab:20s} delta_i/delta_thr={di/dthr:.2f}: max rel. difference R-picture vs delta-picture {rel:.1
        e};"
       f" t_ta = {tta:.3f}/H0" + ("" if np.isfinite(tta) else " (none)"))
    runs.append((lab, tt, dR, dD))
# asymptote of the non-turning shell: delta tends to a constant
lab, tt, dR, dD = runs[2]
```

```
say(f" non-turning shell: 1+delta at t = 4, 6, 8 /H0: " +
    ", ".join(f"{1+run(cases[2][1], tmax=T)[1].y[0][-1]:.3f}" for T in (4, 6, 8)))

# ------------------------------------------------------------------ (iii) nested top hats of the seed
# E = -v^2/2 for every shell: A = GM/v^2, B = GM/v^3; units G = v = 1, outermost mass 1
def eta_of(t, M):
    x = t/M
    if x >= 2*np.pi: return 2*np.pi
    return brentq(lambda e: e - np.sin(e) - x, 0, 2*np.pi)
Ms = np.geomspace(1e-3, 1, 400)

# ------------------------------------------------------------------ figure
fig, ax = plt.subplots(1, 3, figsize=(15, 4.6))
eta = np.linspace(0.05, 1.97*np.pi, 800); tt = _u(eta); tta = np.pi
ax[0].semilogy(tt/tta, one_plus_delta(eta), "-", c="#1f4e79", lw=2.2, label=r"cycloid (exact)")
tn = np.linspace(t0*1.01, t_end, 60)
ax[0].semilogy(tn/tta, 1+sol.sol(tn)[0], "o", ms=3.5, mfc="none", c="#c0504d", label=r"$\delta$ equation,
     numerical")
ax[0].semilogy(tt/tta, 1+3/20*(6*tt)**(2/3), "--", c="0.45", lw=1.2, label=r"linear: $1+\frac{3}{20}(6t/B)
     ^{2/3}$")
for e, s, y in [(np.pi, "turnaround\n$1+\\delta=5.55$", 14), (1.5*np.pi, "halting\n$1+\\delta=147$", 400)]:
    x = (e - np.sin(e))/tta; ax[0].axvline(x, c="0.7", lw=0.8, ls=":")
    ax[0].text(x - 0.03, y, s, fontsize=8, va="center", ha="right")
ax[0].axvline(2, c="0.7", lw=0.8, ls=":"); ax[0].text(1.93, 3e3, "collapse", fontsize=8, ha="right")
ax[0].set(xlabel=r"$t/t_{\rm ta}$", ylabel=r"$1+\langle\delta\rangle$", ylim=(0.8, 1e4), xlim=(0, 2.05))
ax[0].legend(frameon=False, fontsize=8, loc="upper left")
ax[0].set_title("(a) one top hat, Einstein-de Sitter", fontsize=10)
cols = ["#1f4e79", "#c0504d", "#4f8a3c"]
for (lab, tt2, dR, dD), c in zip(runs, cols):
    a2 = a_of_t(tt2)
    ax[1].loglog(a2, 1+dD, "-", c=c, lw=2, label=lab)
    k = np.linspace(0, len(tt2)-1, 40).astype(int)
    ax[1].loglog(a2[k], 1+dR[k], "o", ms=3.2, mfc="none", c=c)
ax[1].axvline(1, c="0.7", lw=0.8, ls=":"); ax[1].text(1.05, 1.3, "today", fontsize=8)
ax[1].set(xlabel="scale factor $a$", ylabel=r"$1+\delta$", ylim=(1, 1e4), xlim=(0.02, 30))
ax[1].legend(frameon=False, fontsize=8, loc="upper left",
           title=r"lines: $\delta$ equation; circles: shell equation", title_fontsize=8)
ax[1].set_title(r"(b) flat $\Lambda$CDM: two pictures, one solution", fontsize=10)
for T, c in zip([0.02, 0.1, 0.4, 1.0], ["#9bbbd9", "#5b8fc0", "#2d5f93", "#0b2e59"]):
    e = np.array([eta_of(T, M) for M in Ms])
    od = np.where(e < 1.5*np.pi, one_plus_delta(np.maximum(e, 1e-4)), np.nan)
    ax[2].loglog(Ms, od, "-", c=c, lw=1.8, label=fr"$t={T}\,GM_{{\rm out}}/v^3$")
    Mb = T/(1.5*np.pi + 1)
    if Mb > Ms[0]: ax[2].axvspan(Ms[0], Mb, color=c, alpha=0.08)
Mlin = np.geomspace(1e-3, 1, 10)
ax[2].loglog(Mlin, 1 + 3/20*(6*0.02/Mlin)**(2/3), "--", c="0.45", lw=1, label=r"linear at $t=0.02$, $\propto
     M^{-2/3}$")
ax[2].set(xlabel=r"shell mass $M/M_{\rm out}$", ylabel=r"$1+\langle\delta\rangle(M,t)$", ylim=(1, 300))
ax[2].legend(frameon=False, fontsize=8, loc="upper left")
ax[2].set_title("(c) the seed: nested top hats (shaded: halted)", fontsize=10)
fig.tight_layout(); fig.savefig("fig_tophat.png", dpi=200)
say("figure: fig_tophat.png"); out.close()
```

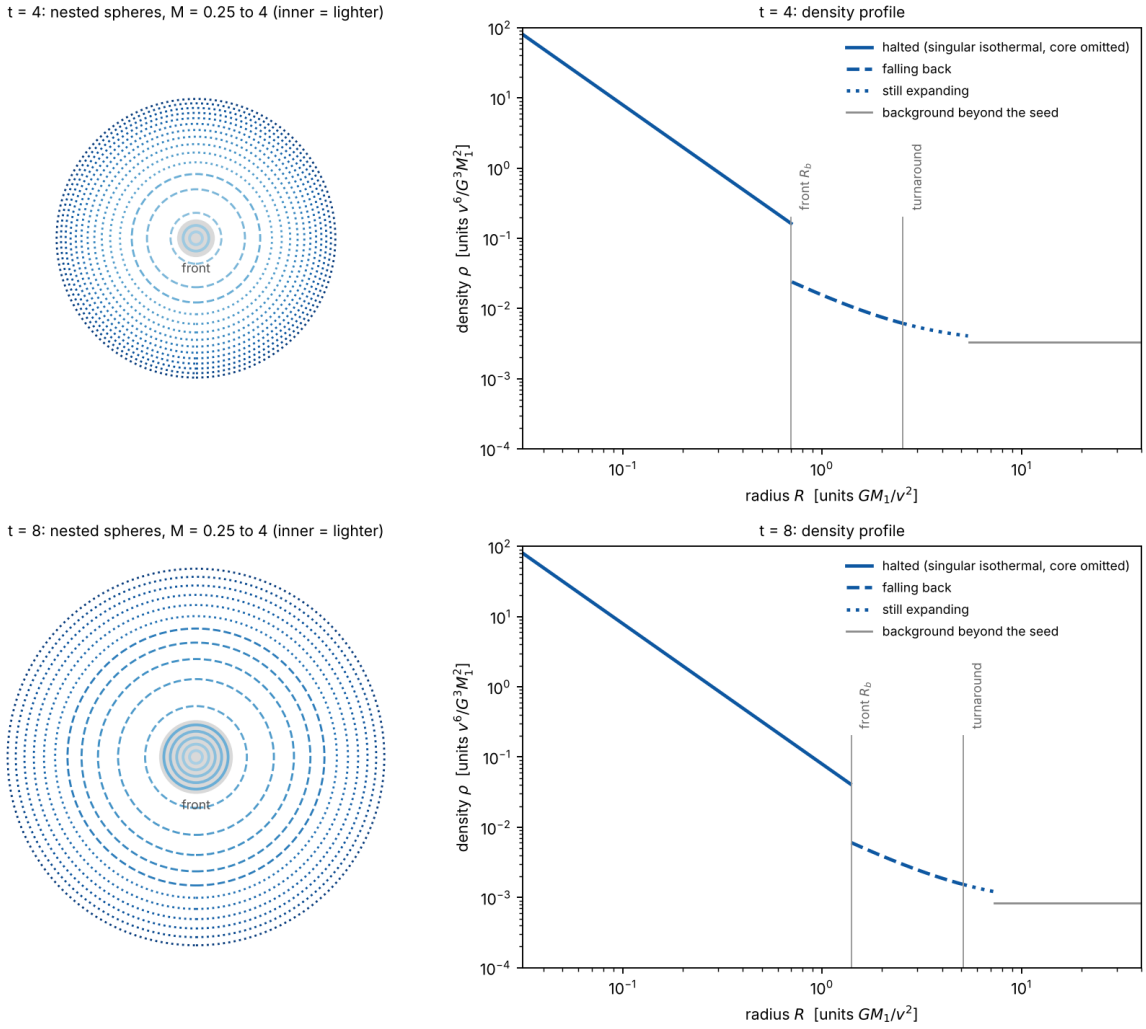


Figure 3: The seed of Fig. 2, extended to $M = 4$, at $t = 4$ (top) and $t = 8$ (bottom); units $G = v = 1$. Left: the nested spheres, $M = 0.25$ to 4 (inner = lighter): halted inside the front (solid, grey disc), falling back (dashed), still expanding (dotted). Right: the density profile at the same time, with the same line styles: the halted matter as a singular isothermal sphere (core omitted), the jump by $\simeq 6.7$ at the front, the infalling and expanding shells, and the background beyond the edge of the seed (grey). Vertical lines mark the front R_b and the current turnaround radius. Script: `code/nested2.py`.